

EDUCATION

B.Sc. in Life Science & B.Eng. in Applied Artificial Intelligence – cum laude Handong Global University (GPA: 4/4.5) Teaching Assistant: AI Application Programming, Big Data Modeling and Platforms	Feb 2023 – Feb 2026 Pohang, South Korea
B.A. in Psychology National Institute for Lifelong Education (GPA 4/4.3)	Feb 2021 – Feb 2023 Seoul, South Korea

RESEARCH EXPERIENCE

Probabilistic Denoising Pipeline for Data Quality Control Handong Global University under Prof. Jaeyoung Chun - Developed a probabilistic diffusion-based denoising pipeline for quality control in single-cell data preprocessing, recovering 26% of previously discarded cells by replacing rigid threshold-based filtering with learned probabilistic representations. - Engineered an upstream data pipeline with high-recall design to preserve minority-class distributions, improving downstream data yield and supporting more robust model training on imbalanced datasets.	Sep 2025 – Dec 2025 Capstone Project
Large-Scale Image Classification Pipeline University of Nevada, Las Vegas, DataX Lab under Prof. Mignon Kang - Engineered a hybrid deep learning pipeline integrating Vision Transformer embeddings with sequential classifiers across a 4-person international research team, achieving 0.99 AUC on large-scale imbalanced classification tasks. - Designed a lightweight patch tiling and filtering algorithm that reduced raw data volume by 80% (128 GB → 25 GB) without sacrificing model-critical features, significantly lowering computational overhead for the distributed compute environment.	Jun 2025 – Jul 2025 Project Lead
Multi-Institution Research Project on NLP Data Pipeline Seoul National University, Center of Convergence Approaches for Drug Development (CCADD) under Prof. Howard Lee - Coordinated cross-institutional communication across 3 research organizations from project kickoff through weekly follow-ups during the research initiation phase. - Supported data standardization for a large-scale ministry-funded project by collecting and organizing metadata examples to establish a structured data management framework.	Jan 2025 – Apr 2025 Undergraduate Researcher
Predictive Modeling & Optimization Pipeline for JUMP AI 2024: AI Drug Development Competition Handong Global University under Prof. Jaeyoung Chun - Led end-to-end regression pipeline development to predict continuous target variables, achieving top 0.3% ranking (3rd of 990 teams) as a sole undergraduate-led team among winners, invited to present methodology at the AI Pharma Korea Conference 2024. - Drove model performance through systematic feature engineering via Particle Swarm Optimization and hyperparameter tuning with Optuna and MLflow, establishing a reproducible experiment tracking workflow.	Aug 2024 – Oct 2024 Project Lead
Sequence-Aware Transformer Preprocessing Framework Handong Global University, BioData Lab under Prof. Taejin Ahn - Designed a transformer-based preprocessing framework that treats sequential data as structured language, achieving 0.71 AUC, eliminating the need for manual feature selection by enabling automated contextual representation learning end-to-end. - Validated that transformer-based embeddings capture latent structure without domain-specific intervention, reducing analyst-dependent variability; published as undergraduate thesis.	Mar 2024 – Nov 2024 Intern, Capstone Project

KEY PROJECTS

Travel Planning Agent Platform 3-person Cross-functional Team - Leading product strategy for an agentic travel-planning platform built with LangGraph, translating user needs into engineering requirements. - Designed scalable user flow and design system from scratch, aligning business logic with technical constraints through rapid prototyping.	Mar 2026 – Present PM, UI/UX, Product Strategy
Jobis: AI Agent Interview Preparation Assistant 2025 Summer Bigdata Camp (5-person team) - Contributed to an AI interview preparation assistant that generates role-specific questions from a user's CV and JD via LangGraph and RAG. - Presented final solution to camp evaluators, communicating system design and user value to a mixed audience.	Summer 2025 Developer [GitHub]
Course Q&A Chatbot 2024 Deep Learning – LLM Study Group - Independently developed a RAG-based chatbot using LangChain, enabling students to query course content via natural language. - Constrained responses to syllabus scope via prompt engineering, reducing hallucination for domain-specific queries.	Fall 2024 Developer [GitHub]

HONORS AND AWARDS

Overseas Research Excellence Scholarship Awarded by the Big Data Consortium for outstanding participation at University of Nevada, Las Vegas	Summer 2025
3rd place, Daewoong Pharmaceutical President's Award Ranked 3rd out of 990 teams in <i>JUMP AI 2024: AI Drug Discovery Competition</i> ; Sole undergraduate-led team among winners	Oct 2024
Big Data Seagull Scholarship Merit-based scholarship for academic performance in the curriculum certified by HGU School of Applied Artificial Intelligence	Spring 2024, Fall 2024

CONFERENCES AND PRESENTATIONS

3. No Cell Left Behind: Recovering Discarded Cells via Self-Supervised Diffusion Denoising Handong Global University Bio-AI Capstone Design Fair	Dec 2025 Oral Presentation
2. Diagnosing Cancer with Genomic Context Handong Global University Capstone Design in Life Science Fair	Nov 2024 Oral Presentation
1. IRAK4 IC50 Inhibition Prediction AI Pharma Korea Conference 2024	Oct 2024 Poster Presentation

TEACHING EXPERIENCE

Introduction to AI and Machine Learning, Teaching Assistant Addis Ababa Science and Technology University (Handong UNESCO-UNITWIN Program) <u>Instructor:</u> Prof. Dai Yon Cho	Aug 2025 Addis Ababa, Ethiopia
Big Data Modeling and Platforms, Teaching Assistant Handong Global University, School of Applied Artificial Intelligence <u>Instructor:</u> Prof. Jaeyoung Chun	Mar 2025 – Jun 2025 Pohang, South Korea
AI Application Programming, Teaching Assistant Handong Global University, School of Mechanical and Control Engineering <u>Instructor:</u> Prof. Sangsan Lee	Mar 2025 – Jun 2025 Pohang, South Korea

COMMUNITY INVOLVEMENT

R Programming Camp, Teaching Assistant Mentored high school students in data analysis and visualization using R at Handong Global University	Jan 2026 Pohang, South Korea
Handong Office of International Affairs, Exchange Student Mentor Assisted exchange students with adaptation to Korean culture and Residential College (RC) life	Aug 2024 – Dec 2024 Pohang, South Korea
Python Programming Camp, Teaching Assistant Provided assistance for elementary to high school students through hands-on sessions at Incheon Beauty and Art High School	Aug 2023 Incheon, South Korea